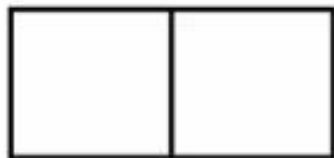


VÁLVULAS



SENAÍ

Número de vias/número de posições vias



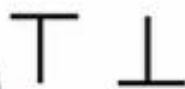
O número de quadrados representados no símbolo é igual ao número de posições da válvula.



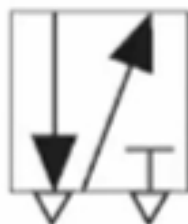
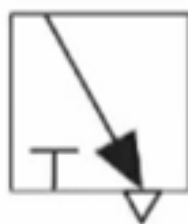
O número de vias de trabalho (entradas e saídas) é representado por linhas na parte externa do símbolo.



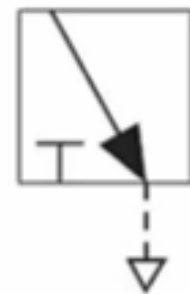
As setas indicam a direção do fluxo de ar.



Conexões de bloqueio são identificadas em ângulos retos.



Escape Interno

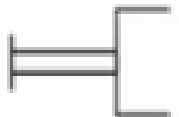
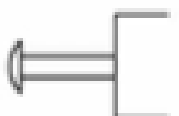


Escape Externo



Identificação das Vias	DIN ISO 5599-3	Sistema de Letras
Via de Pressão	1	P
Vias de Trabalho	2 , 4 , 6	A , B , C
Vias de Exaustão (Escape)	3 , 5 , 7	R , S , T
Vias de Dreno (apenas hidráulica)	- - -	L
Vias de Pilotagem	10 , 12 , 14	X , Y , Z



Acionamento Manual	
Geral (manutenção)	
Botão	
Alavanca	
Pedal	



Acionamento Mecânico	
Came	
Rolete	
Rolete Escamoteável	
Mola	
Centrada por Mola (válvulas com 3 posições)	



Acionamento Pneumático

Acionamento Direto
Simples Piloto



Acionamento Direto
Duplo Piloto



Acionamento Indireto
Simples Servo-Piloto



Acionamento Indireto
Duplo Servo-Piloto



Acionamento Elétrico

Simples Solenoide



Duplo Solenoide



Acionamentos Combinados

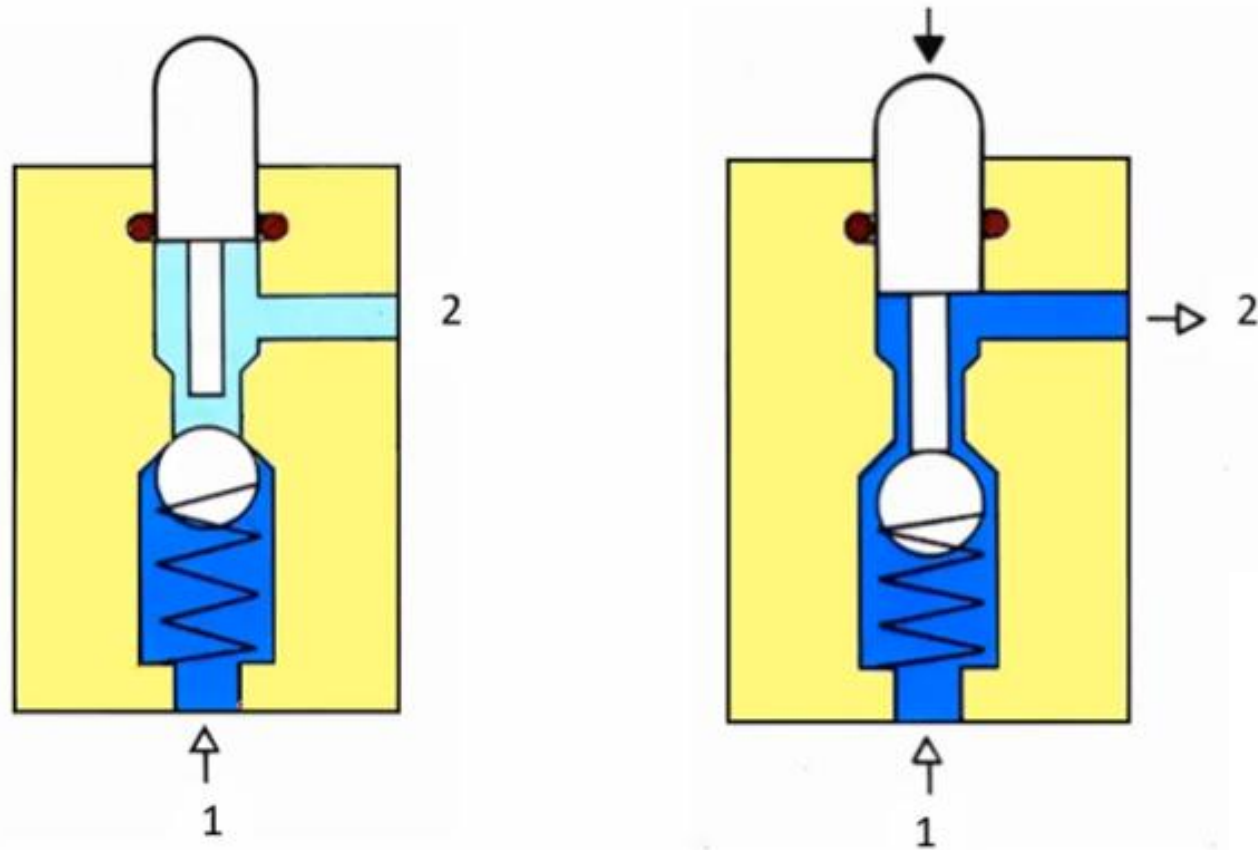
Duplo Servo-Solenoide
com acionamento manual
de manutenção





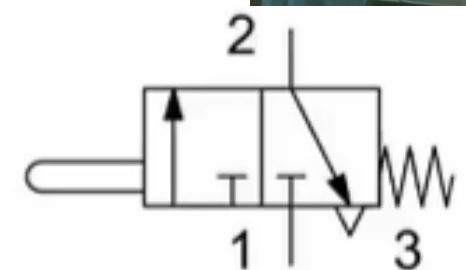
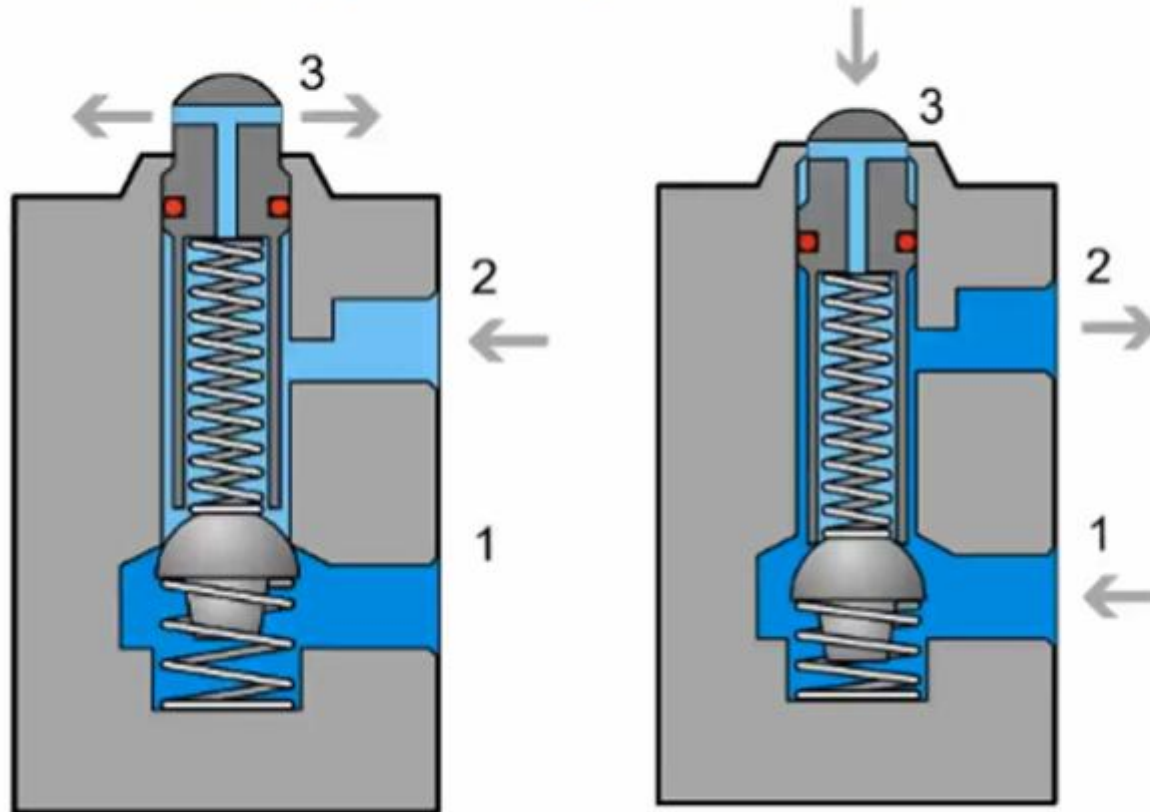
ção

Válvula 2/2 vias, Normalmente Fechada



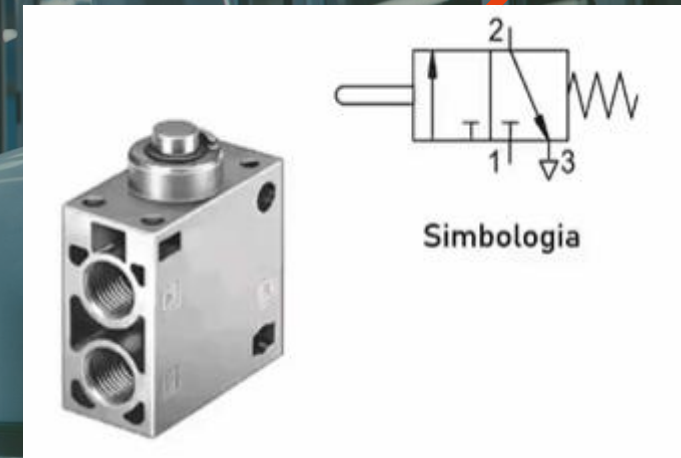
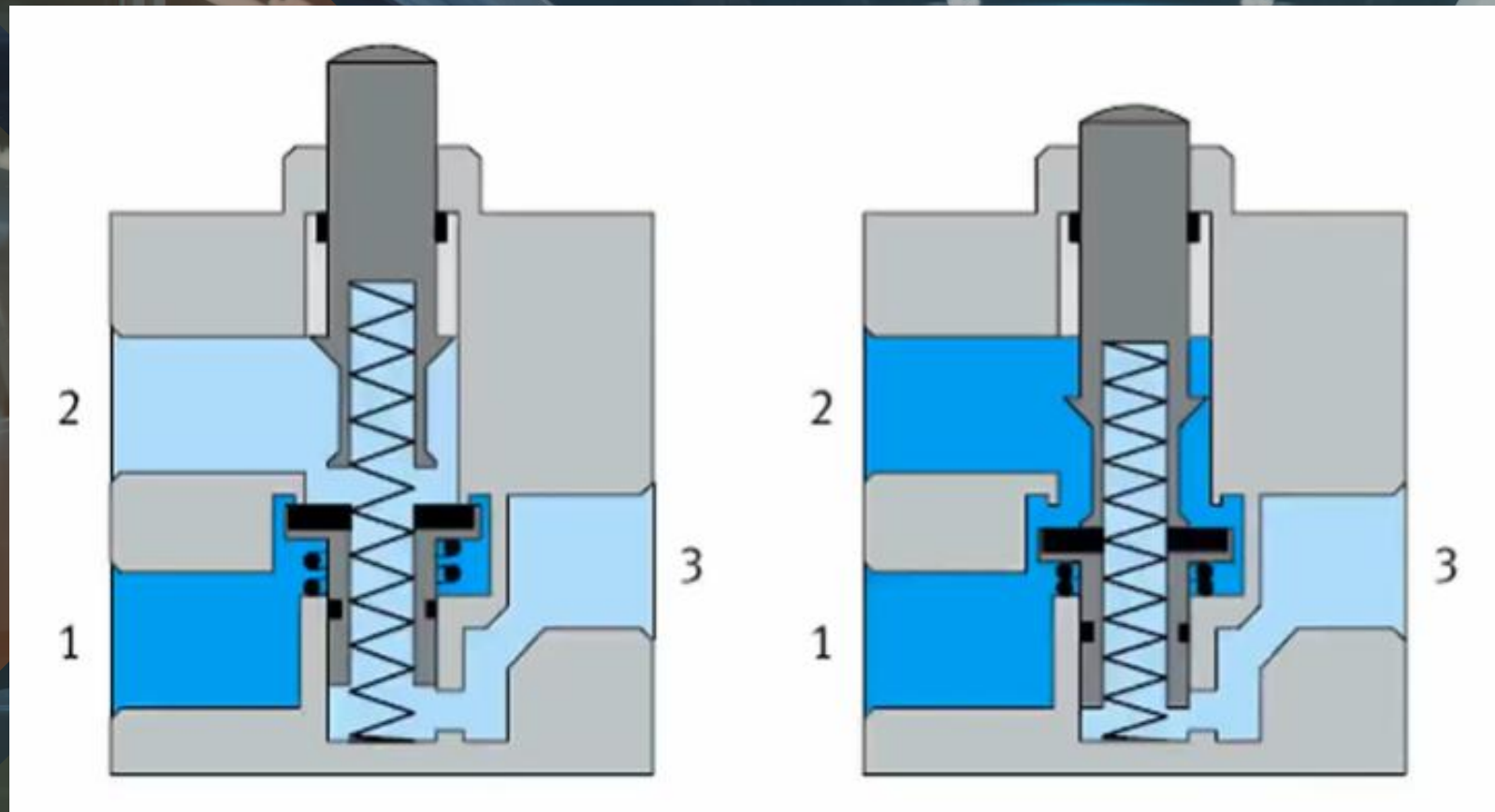
Válvula 3/2 vias, normalmente fechada, acionada por came, retorno por mola

Válvula 3/2 vias, Normalmente Fechada



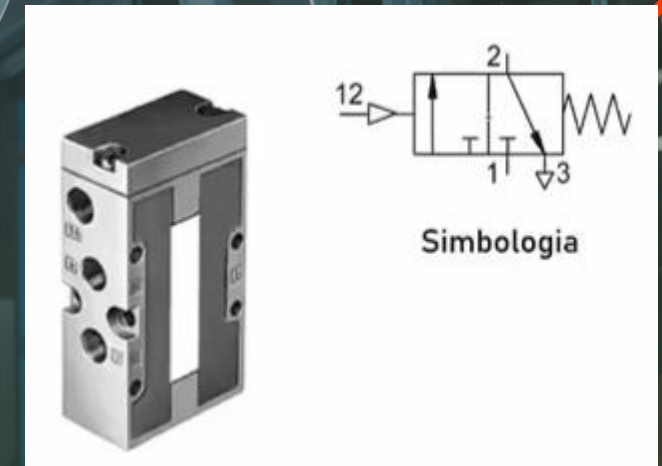
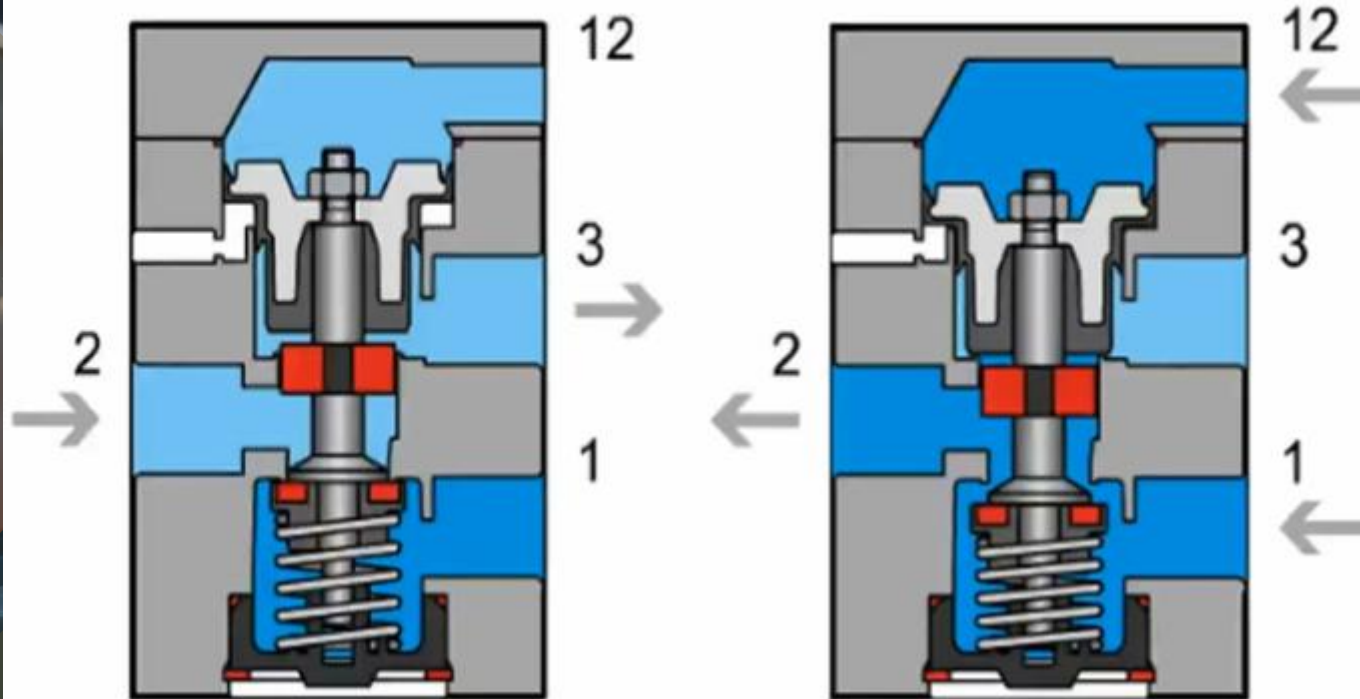
Simbologia

Válvula 3/2 vias, normalmente fechada, acionada por came, retorno por mola



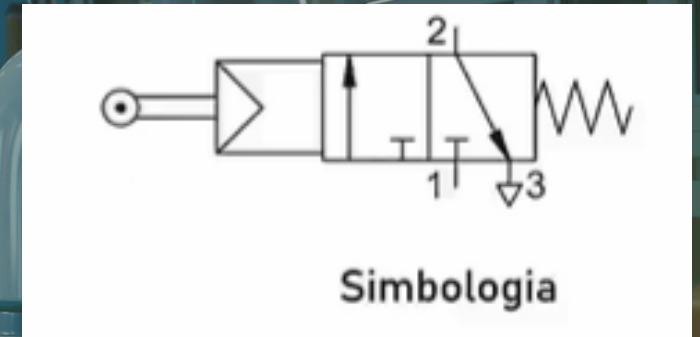
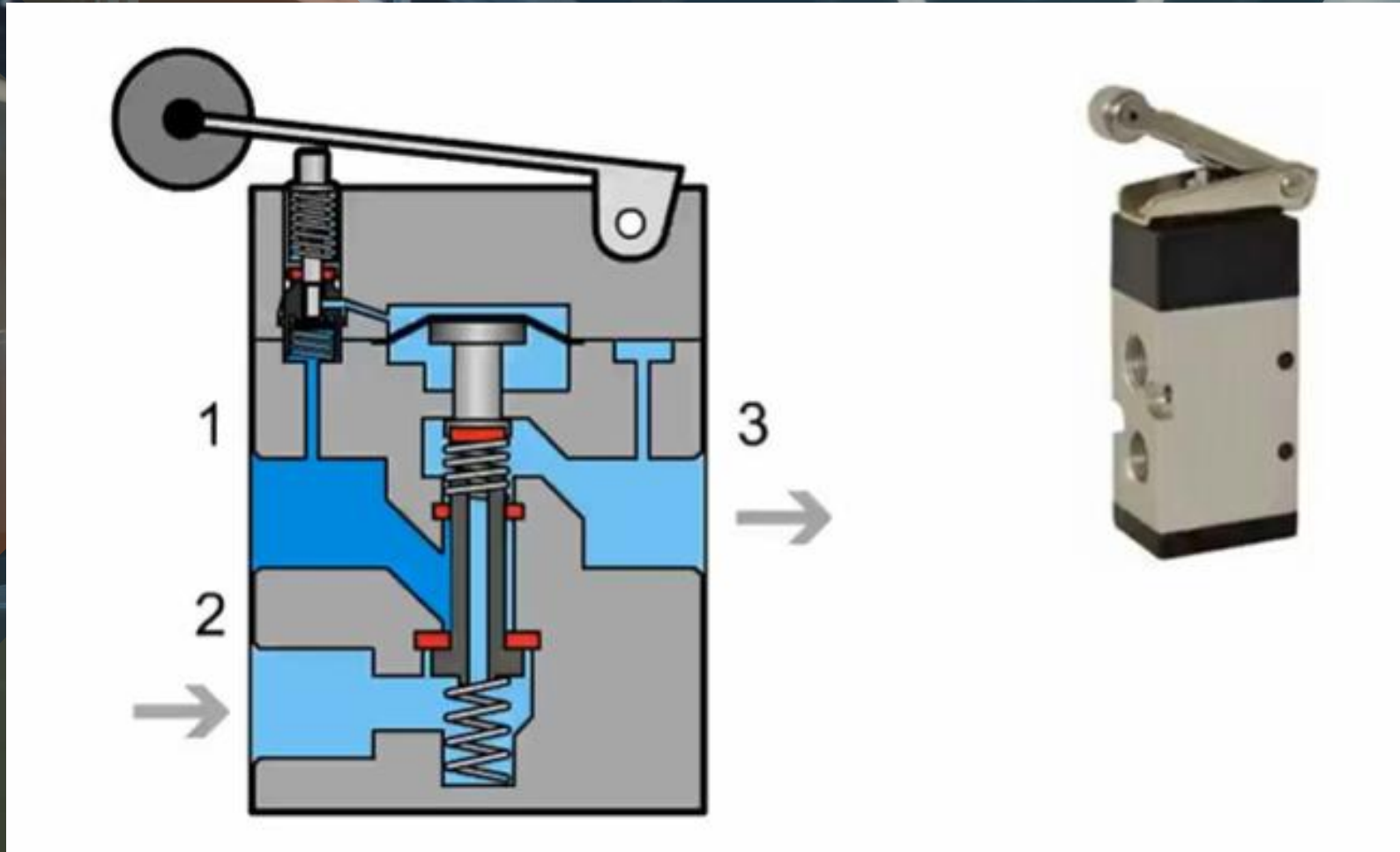


Válvula 3/2 vias, normalmente fechada, acionada por simples piloto



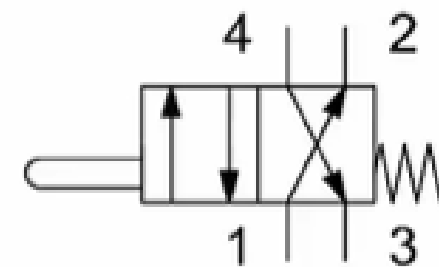
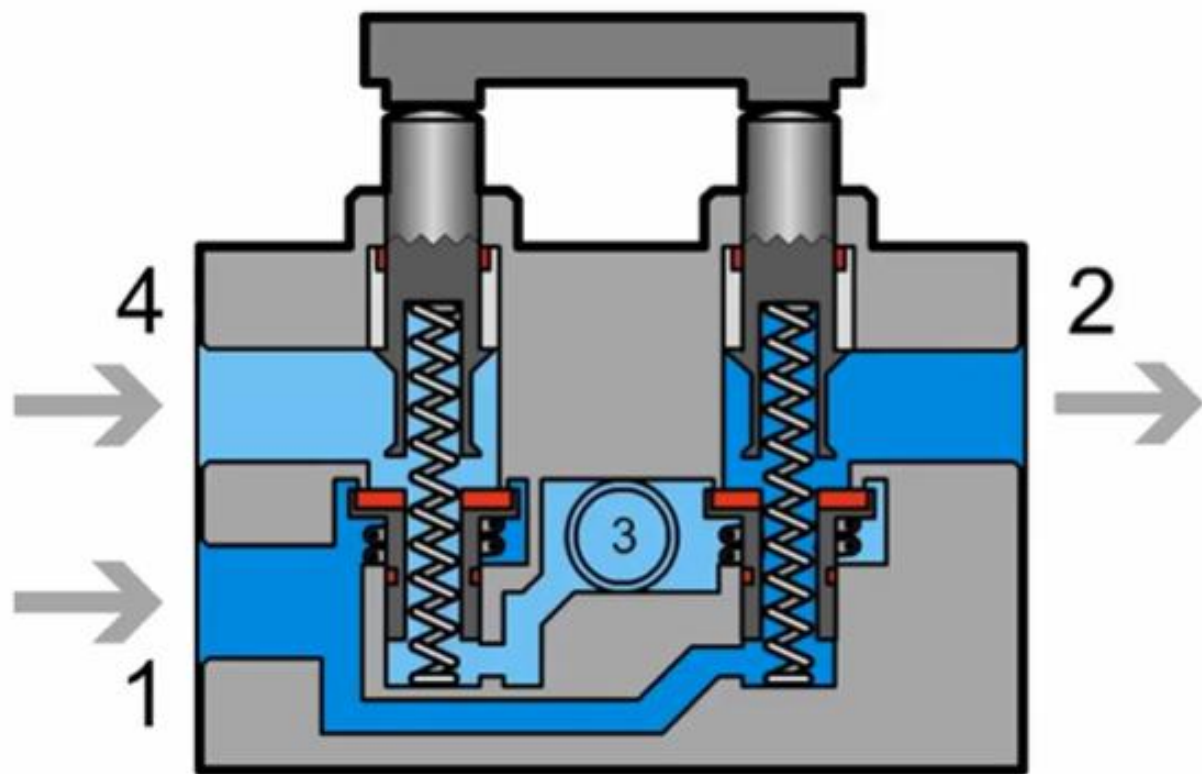


Válvula 3/2 vias, normalmente fechada, acionada por rolete, retorno por mola





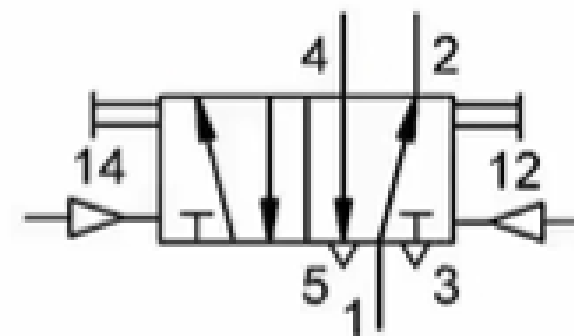
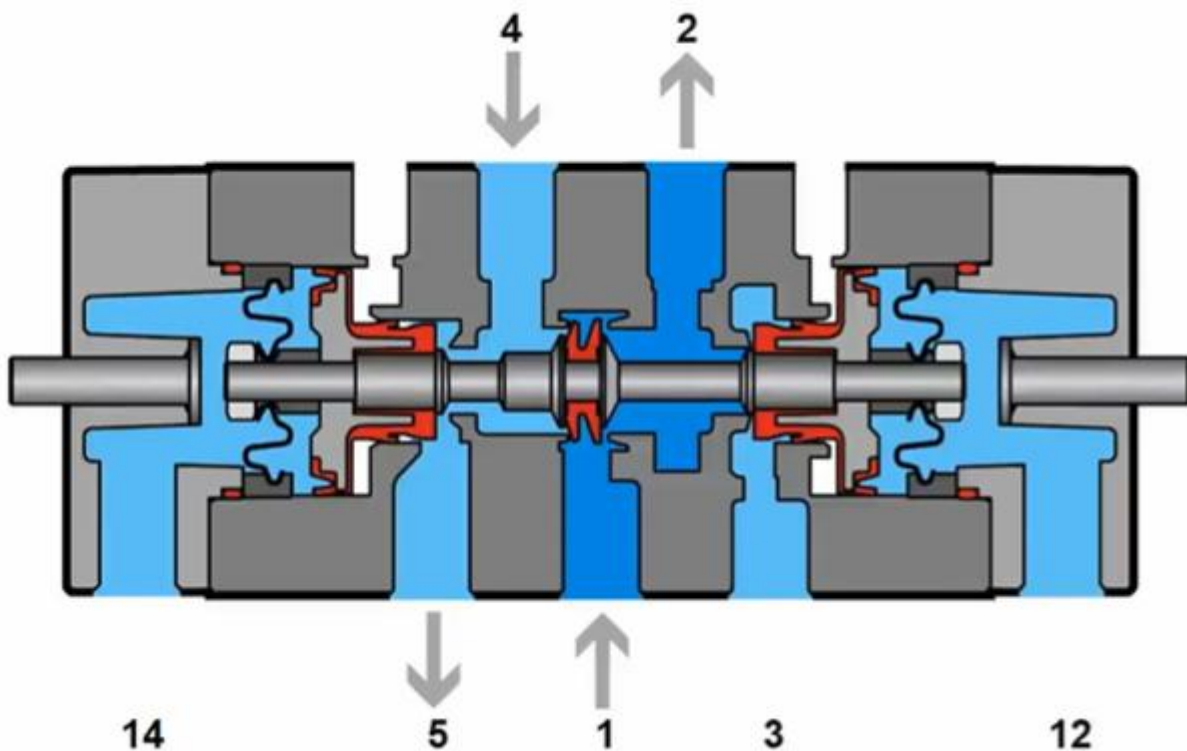
Válvula 4/2 vias, acionada por came, retorno por mola



Simbologia



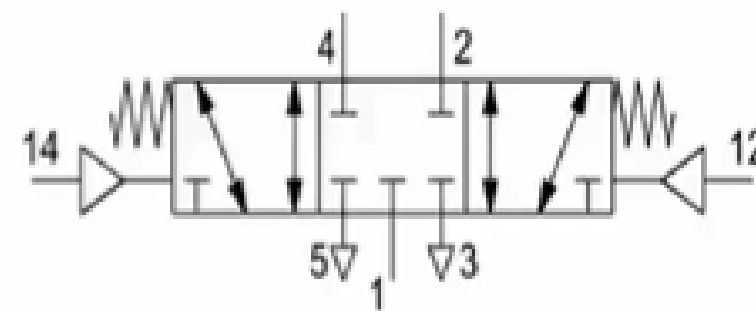
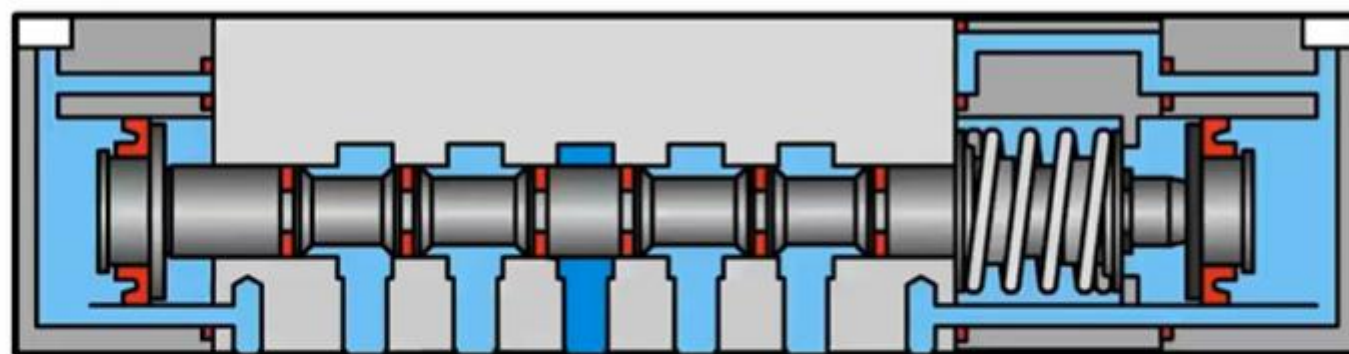
Válvula 5/2 vias, duplo piloto



Simbologia



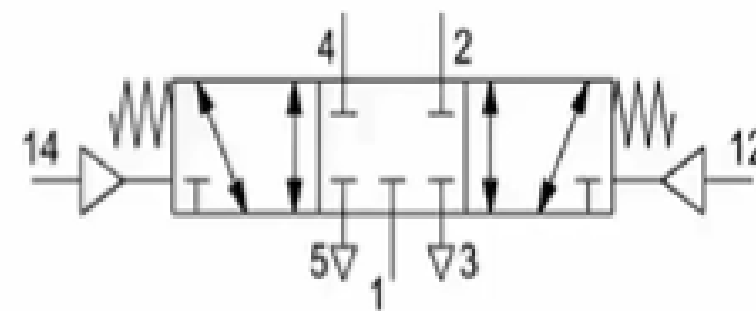
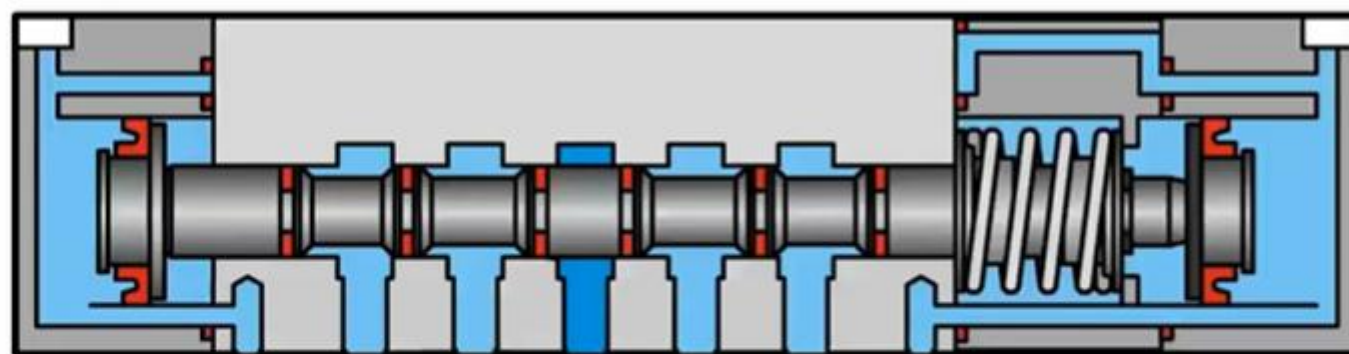
Válvula 5/2 vias, duplo piloto, centro fechado



Simbologia



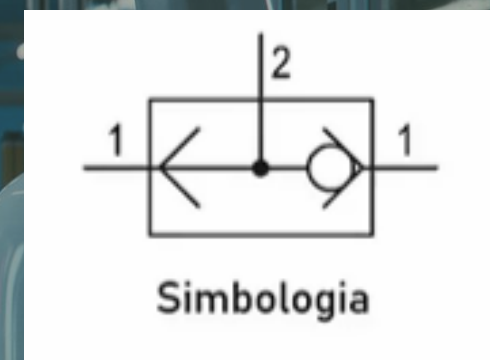
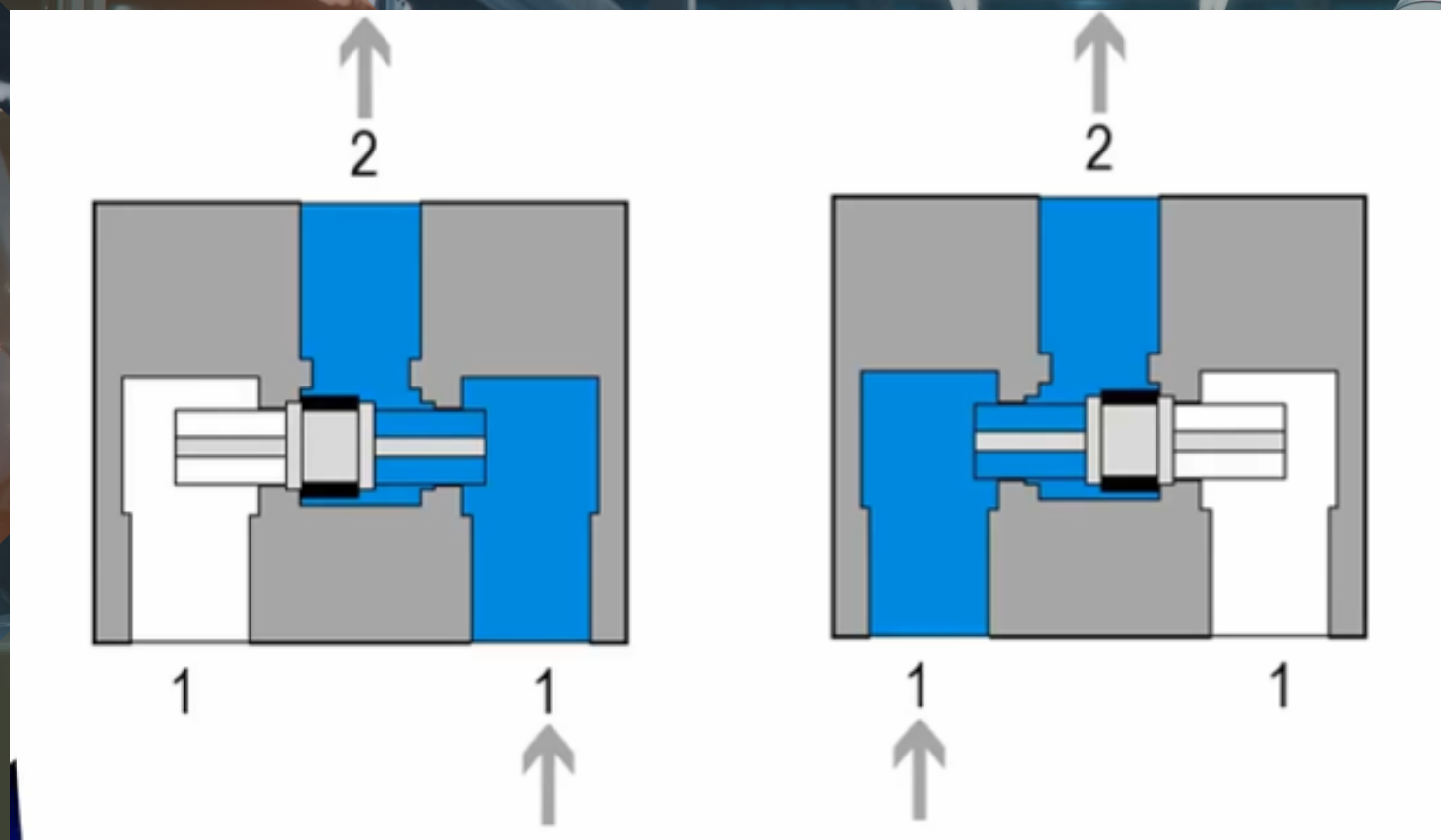
Válvula 5/2 vias, duplo piloto, centro fechado



Simbologia

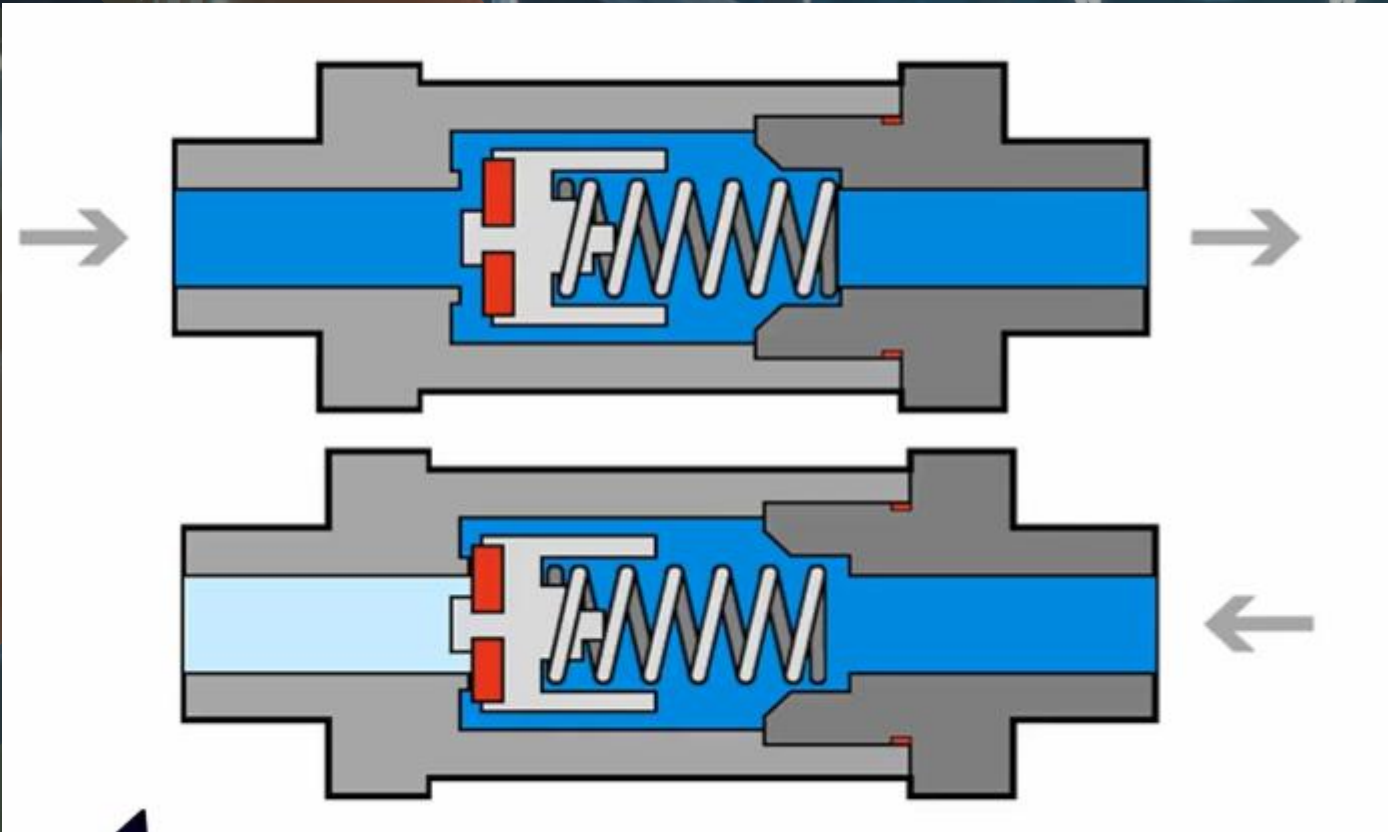


Válvula alternadora (válvula OU)



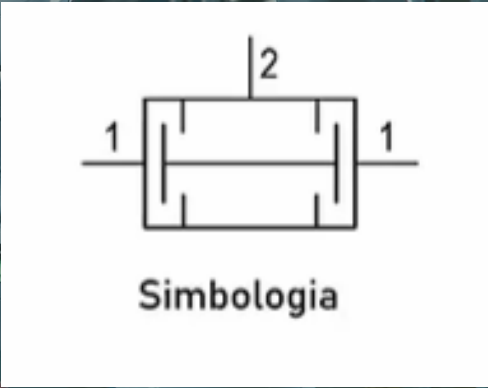
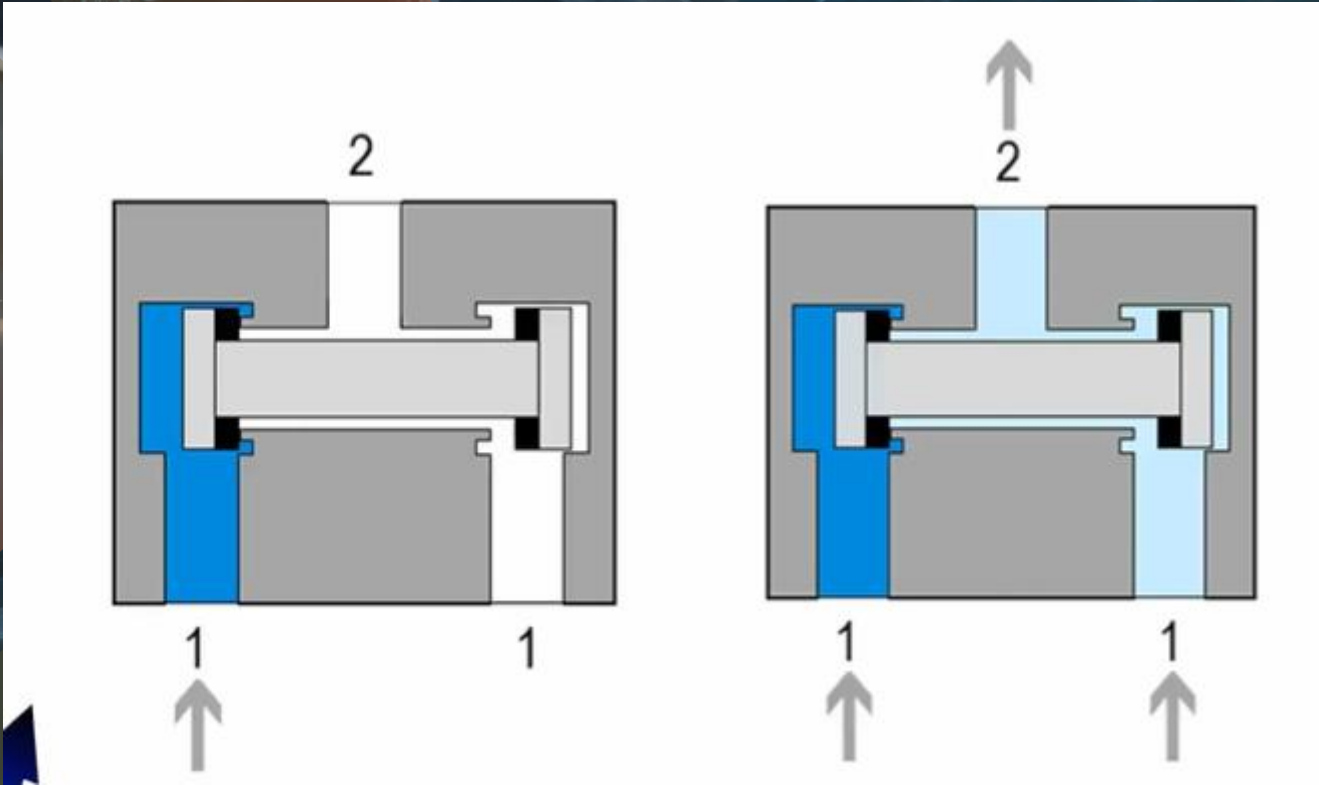


Válvula de retenção



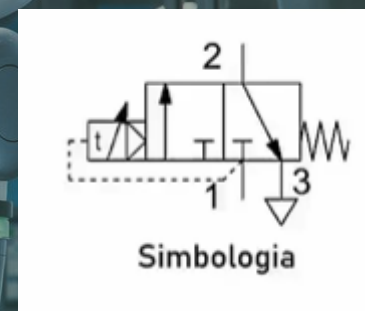
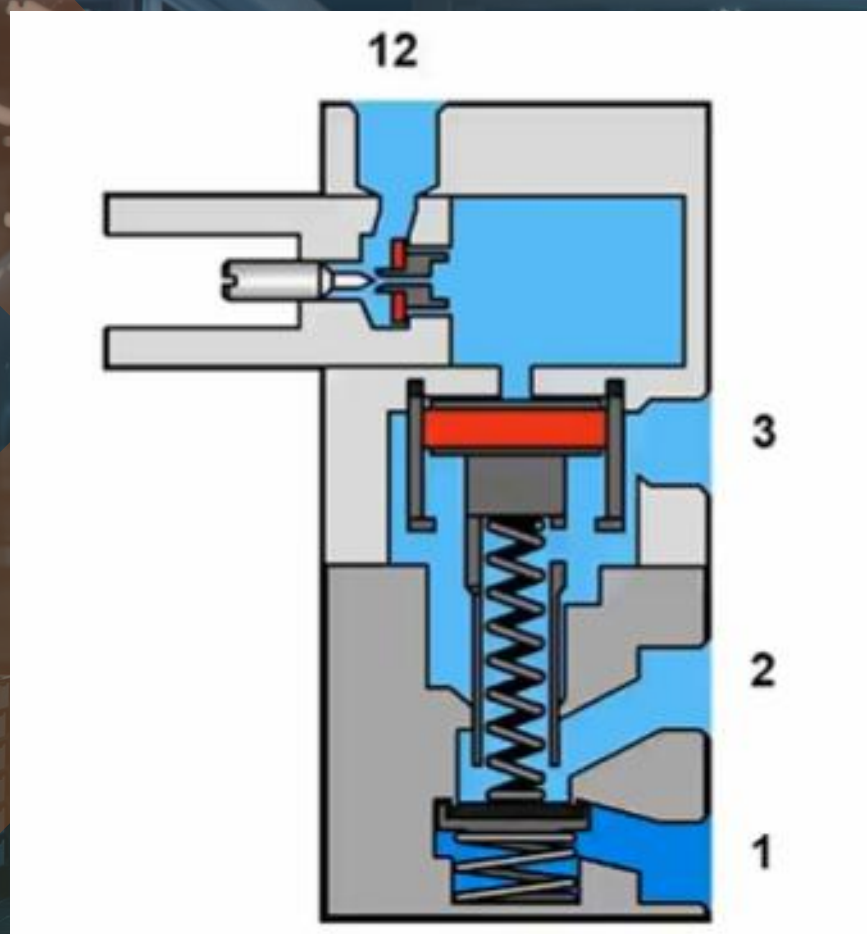


Válvula de simultaneidade (e)



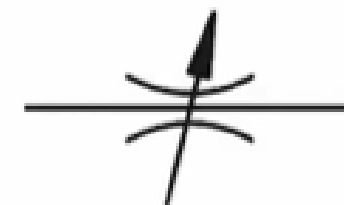
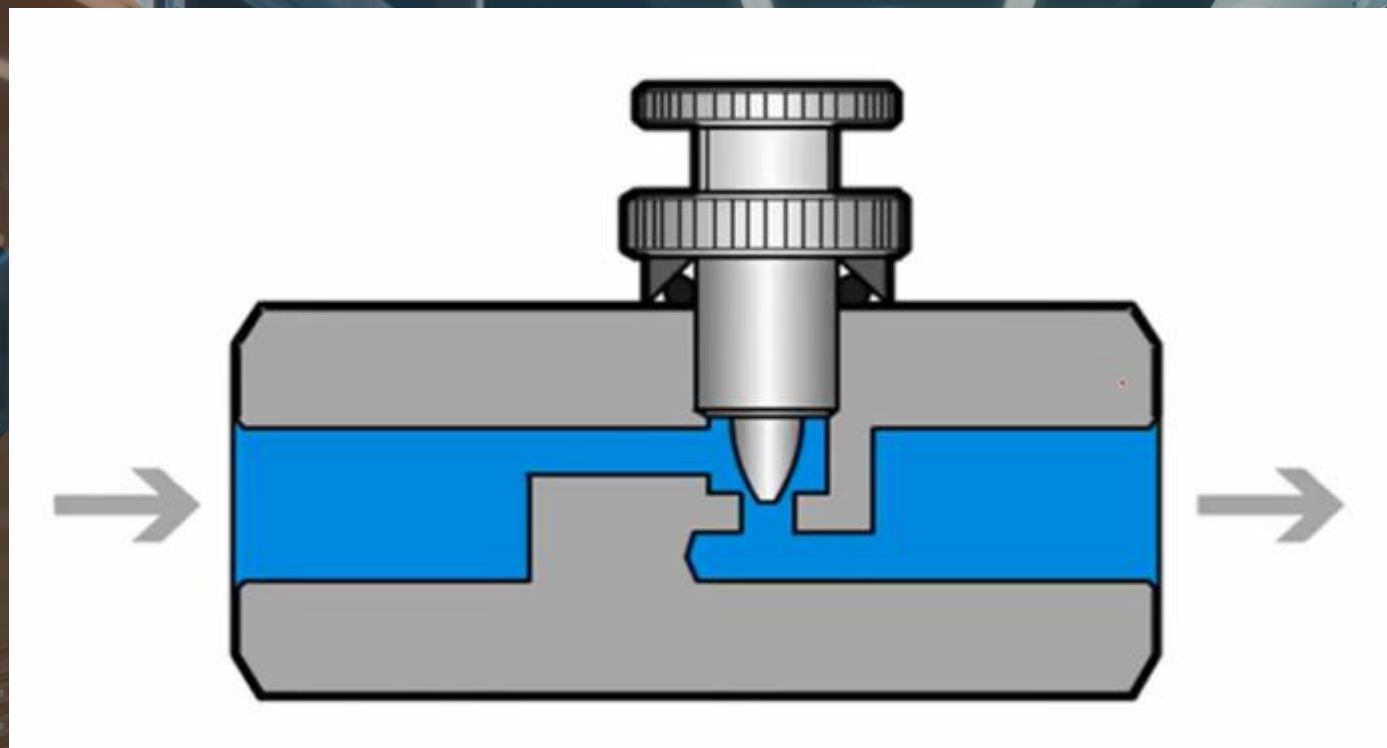


Válvula temporizadora



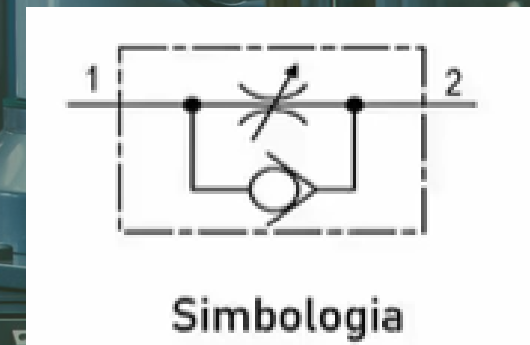
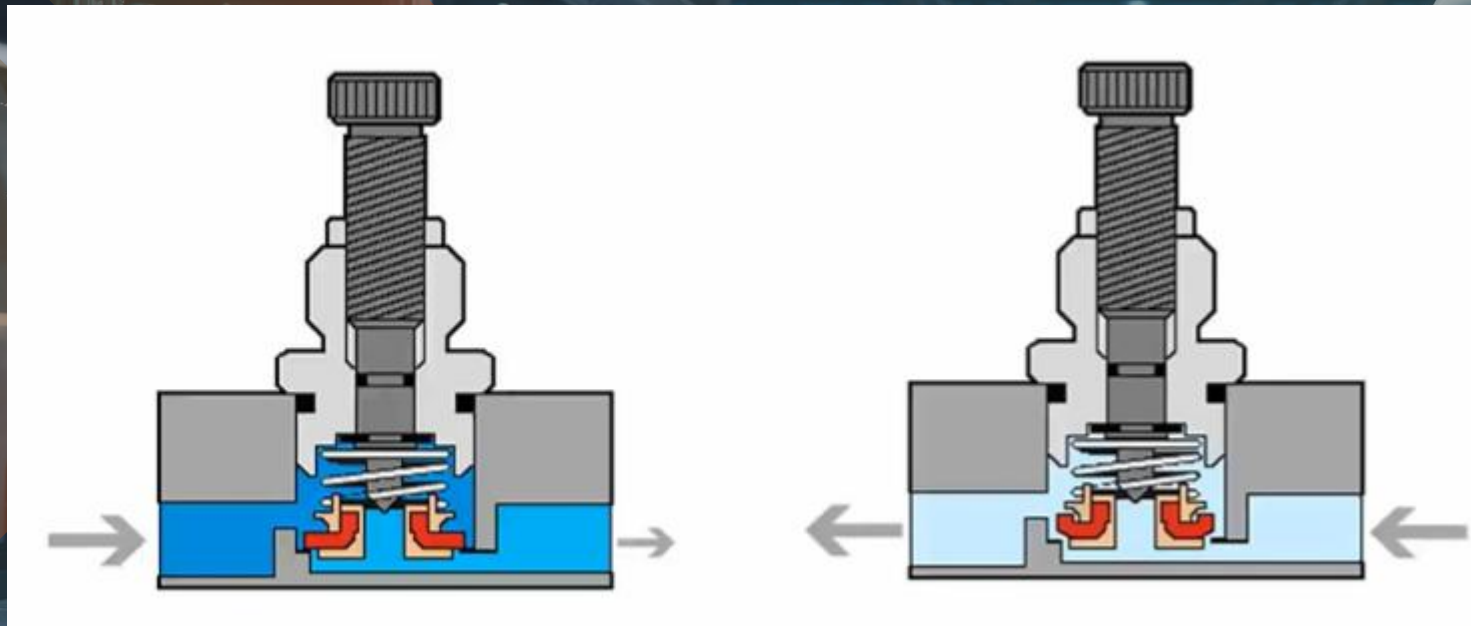


Válvula de controle bidirecional



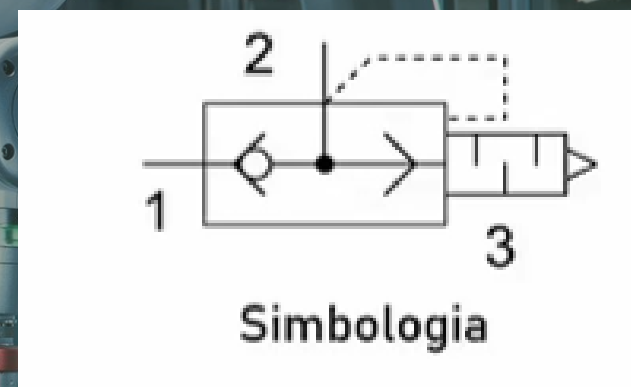
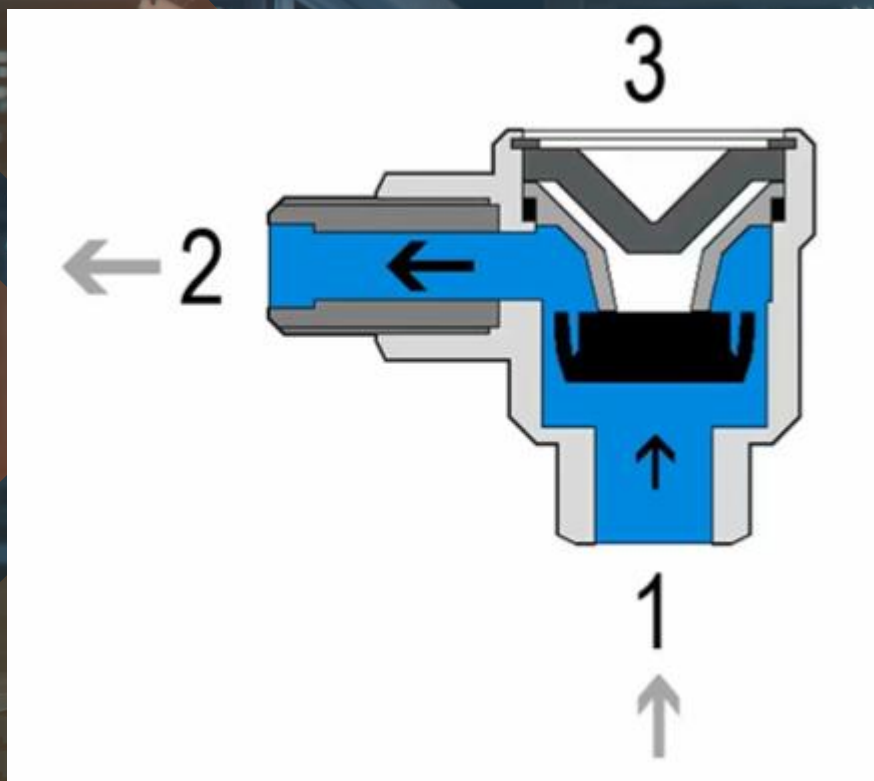
Simbologia

Válvula de controle unidirecional





Válvula de escape rápido



Obrigado!
SENAI

